BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: BN-XIIXXII-I1 / BN-XII-XXII-II

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I (FRESH)

SECTION–A (Version-A)

Time Allowed: 20 Min | **Maximum Marks:** 20

Note: Q.No.1. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q. If f'(x) = 0 for all 'x' then f(x) is _____

(A) Maximum
(B) Inflection
(C) Minimum
(D) None of these

Q. lim (1+1/x)^x = _____ (n→∞)

(A) x
(B) ∞
(C) 0
(D) e

Q. d/dx ($4x^3$) = _____

(A) $3x^2$
(B) $4x^2$
(C) $12x^2$
(D) 12x

Q. Geometrically derivative represent slope of _____ line to the curve.

(A) Tangent
(B) Normal
(C) Secant
(D) Chord

Q. A line which passes through focus and perpendicular to major axis is called _____

(A) Directrix
(B) Latus Rectum
(C) Focus
(D) Chord

Q. The function f(x) is said to be increasing on interval [a,b], if _____

(A) f'(x)<0
(B) f'(x)=0
(C) f'(x)≠0
(D) f'(x)>0

Q. The point where the convacity of differential function changes is called _____

(A) Inflection point
(B) Critical point
(C) Minima
(D) Maxima

Q. $\int x^2\,dx$ = _____

(A) x^3+C
(B) x^2+C
(C) $3x^2+C$
(D) $x^3/3+C$

Q. $\int_a^a f(x)dx$ = _____

(A) 2a
(B) 0
(C) f(a)
(D) a

Q. The set of all points whose distance from a fixed point is equal to distance from a fixed line is called _____

(A) Hyperbola
(B) Parabola
(C) Ellipse
(D) Circle

Q. The degree of the equation of parabola is _____

(A) 2
(B) -2
(C) 3
(D) 0

Q. The x-intercept of line 2x+4y+6=0 _____

(A) -3
(B) 2
(C) 4
(D) 6

Q. The lines represented by homogenous equation $ax^2+2hxy+by^2=0$ are real and coincident if _____

(A) $h^2-ab>0$
(B) $h^2-ab<0$
(C) $h^2-ab=0$
(D) $h^2-ab\neq0$

Q. The line y=mx+c is tangent to circle $x^2+y^2=a^2$ if c= _____

(A) $\pm\sqrt{(1+m^2)}$
(B) $\pm m\sqrt{(1+a^2)}$
(C) $\pm\sqrt{(1+a^2)}$
(D) $\pm a\sqrt{(1+m^2)}$

Q. The length of tangent from point P(3,4) on circle $x^2+y^2=9$ is:

(A) 9
(B) 4
(C) -4
(D) 7

Q. The length of major axis of ellipse $x^2/9+y^2/4=1$ is _____

(A) 3 (B) 2

(C) 13 (D) 6

Q. The Hyperbola whose asymptotes are at right angle is called _____

(A) Horizontal (B) Vertical

(C) Rectangular (D) Imaginary

Q. The order of differential equation dy/dx = xy is :_____

(A) 1 (B) 2

(C) 3 (D) 4

Q. $f(x,y)=x^3y+xy^3$, then fx = _____

(A) $3x^2+y^3$ (B) $3x^2y+y^3$

(C) x^3+y^2 (D) x^2+2xy

Q. The function y=f(x) has at least one root in [a,b] if _____

(A) f(a) = f(b) (B) f(a) < f(b)

(C) f(a).f(b)>0 (D) f(a).f(b)<0

SECTION–B

Maximum Marks: 50

Q2. (Marks 50) Attempt any TEN parts of the following. All parts carry equal marks.

i. If f(x) = 4x+1 and $g(x) = 2x^2+5x$, then find fog and gof.

ii. Evaluate $\lim(x\to0) (1-\sin x)/(\cos^3 x)$

iii. Find derivative of f(x) = 7/x, using first principle.

iv. If $y = x^2$ e^{sinx}, then find dy/dx

v. Evaluate $\int (3x^2-7)/(x^3-7x)^4\, dx$

vi. Use Maclaurin series to approximate the value of f(x) = sinx at point $x_0=0$

vii. $\vec{r}(t) = (t+t^2)i + 2tk$

viii. Evaluate $\int dx/\sqrt{(a^2-2x^2)}$

ix. Use the quotient rule to find out the derivative of the following function y = (3x−5)/(x−4)

x. Determine the equation of the circle having center at point C(−9,−6) and circumference passes through origin.

xi. Find the composite functions f[g(x)] and g[f(x)] of the functions f(x) = sin x and g(x) = 2x+3.

xii. Show that y = e^x + e^{2x} is solution of $d^2y/dx^2 - 3\ dy/dx + 2y = 0$

xiii. Compute four iterates of bisection method for f(x) = e^x−2−x in [1,1.8]

SECTION–C

Maximum Marks: 30

Note: Note: Attempt any THREE of the following. All questions carry equal marks.

Q3. Q.No: 3.

(a). Find the equation of tangent line to the curve at the indicated value of x. y = sin (2x+π), x = 0.

(b). Evaluate $\int (3x^2+2x-1)/(x(x-1))\ dx$

Q4. Q.No: 4.

(a). Show that lines x+4y+3=0, and 2x+2y+l=0 are concurrent, then find out the point of concurrency.

(b). Evaluate lim(x→0) (secx−1)/(x secx)

Q5. Q.No: 5.

(a). Find the equation of tangent to the ellipse $x^2/9+y^2/4=1$ which is perpendicular to the line 9x+8y−36=0

(b). Determine equation of orthogonal trajectory y=Cxe^x

Q6. Q.No: 6.

(a). If z = xy f(x/y), then show that $x\ \partial z/\partial x + y\ \partial z/\partial y = 2z$

(b). Use Simpson rule to approximate the value of $\int(0 \text{ to } 1)\ x^4dx$ n=3

GROUP – II (RE-APPEAR)

SECTION-A (VERSION A)

Time Allowed: 20 minutes | **Maximum Marks:** 20

Note: Q.No.1. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number in provided bubble sheet.

> Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q. If $f(x) = 3x^2+7x-5$ then f(3) is equal to:

(A) 43
(B) 45
(C) 46
(D) 48

Q. $\lim(\theta\to0) \sin5\theta/\theta$ is equal to:

(A) 0
(B) 1
(C) 5
(D) 1/2

Q. log(a) x is not defined if:

(A) x<0
(B) x>0
(C) x=a
(D) x<a

Q. d/dx ((e^x−e^{−x})/2) = ...?

(A) cosh x
(B) sinh x
(C) tanh x
(D) sech x

Q. In the interval [0,π], the function f(x) = sin x has a maximum at:

(A) 0
(B) π/3
(C) π/4
(D) π/2

Q. If ∇ is the velocity of a moving object, then ∇/|∇| describes its:

(A) Position vector
(B) Acceleration
(C) Speed
(D) Direction of motion

Q. ∫(−3 to 3) |x|dx is equal to:

(A) 3
(B) 9
(C) 18
(D) None of these

Q. Function f(x) is increasing if:

(A) f'(x) = 0
(B) f'(x) > 0
(C) f'(x) < 0
(D) None of these

Q. Distance between points P(3,−2) and Q(−1,−5) is:

(A) 0
(B) 1
(C) 3
(D) 5

Q. The equation of the line with slope 4, when the y-intercept is 6 is:

(A) y = 6x+4
(B) y = 4x+6
(C) x+y = 6
(D) x−y = 4

Q. The perpendicular distance from a line $7x+3y-9=0$ to a point P(2,3) is:

(A) 14/58 (B) −14

(C) $14/\sqrt{58}$ (D) 58/14

Q. The vector equation of x−axis is:

(A) $\vec{r}\cdot\vec{i} = 0$ (B) $\vec{r}\cdot\vec{j} = 0$

(C) $\vec{r}\cdot\vec{i} = x$ (D) $\vec{r}\cdot\vec{j} = y$

Q. If $3x-8y-7 = 0$, then point $(-3,-1)$ lies:

(A) On the line (B) Above the line

(C) Below the line (D) Opposite to the line

Q. The distance of P(x,y) from the origin is:

(A) $x+y$ (B) x^2+y^2

(C) $\sqrt{(x^2+y^2)}$ (D) None of these

Q. If the eccentricity e = 5/3, then conic will be a:_____

(A) Circle (B) Parabola

(C) Ellipse (D) Hyperbola

Q. The degree of differential equation $(d^2y/dx^3)^2 - d^3y/dx^2 + dy/dx + 5 = 0$ is:

(A) 5 (B) 0

(C) 2 (D) 1

Q. The specialty of Euler's theorem is to verify the degree of a:

(A) Linear function (B) General function

(C) Homogenous function (D) None of these

Q. The equation of a circle with radius 4 units is:

(A) $x^2+y^2 = 2$ (B) $x^2+y^2 = 16$

(C) $x^2+y^2 = 4$ (D) $x^2+y^2 = 8$

Q. The function $(\sqrt{x}+\sqrt{y})/(y+x)$ is homogenous of degree:

(A) 1 (B) 0

(C) 1/2 (D) No degree

Q. The process of shifting of origin only where the new axes are parallel to the old one is called:

(A) Translation (B) Rotation

(C) Transformation (D) Linearization

SECTION-B

Maximum Marks: 50

Q2. (Marks 50) Attempt any TEN parts of the following. All parts carry equal marks.

i. Determine the inverse function of y = f(x) = (x+4)/2.

ii. Evaluate lim(x→25) (√x−5)/(x−25)

iii. Differentiate y = x^n by first principle rule.

iv. Find dy/dx, when y = (cosx)^{logx}.

v. Evaluate ∫(1 to 4) (lnx)/x^3 dx.

vi. Use Taylor's series to approximate the value of a function f(x) = e^x at a point x_0 = 2.

vii. Find the perpendicular distance from a line 7x+3y−9 = 0 to a point P(2,3).

viii. Find an equation of a circle which contains the point (2,6), (2,3) and has a center on the line 3x+2y−1 = 0.

ix. If $\vec{v}$ = 2$\vec{i}$−$\vec{j}$+5$\vec{k}$ and $\vec{w}$ = $\vec{i}$+2$\vec{j}$−3$\vec{k}$ are the two vector functions, then evaluate d/dt ($\vec{v}$+t $\vec{w}$).

x. Find the normal equation at the point P(2,−4) to parabola y^2 = 8x.

xi. Solve the initial value problem dy/dx = cosx, y(0) = 1.

xii. Show that the function f(x,y) = 2xy+y^2 is a homogenous function of degree 2.

xiii. Using trapezoidal rule, evaluate ∫(−1 to 3) x^3dx with n = 4

SECTION-C

Maximum Marks: 30

Note: Note: Attempt any (THREE) questions of the following. All questions carry equal marks.

Q03. Q.NO.03.

a) Evaluate lim(x→0) tanx/x

b) Differentiate y = 7/x by first principle.

Q04. Q.NO.04.

a) Evaluate the integral ∫ dx/(x√(x^4+4))

b) Find a joint equation of the straight line that passes through the origin and perpendicular to the lines represented by 3x^2−7xy+2y^2 = 0.

Q05. Q.NO.05.

a) Find tangent and normal equations at a point (1,2) to the circle x^2+y^2 = 5.

b) For what value of c, the line $x-y+c = 0$ will touch the ellipse $x^2/4+y^2/9 = 1$.

Q06. Q.NO.06.

a) Find the eccentricity and distance between the foci of the ellipse $x^2/4+y^2/9 = 1$.

b) If $u = \tan^{-1} (x^2+y^2)/(x+y)$, then show that $x\ \partial u/\partial x + y\ \partial u/\partial y = 10u$.